



The government is introducing new technologies and climate-resistant variants of crops; it has also placed export curbs to stop price fluctuation



It has brought in policies to boost vegetable cultivation, expand production bases to more states



However, longer-term solutions for mitigation remain elusive

globally, and August was the driest on record in India, he notes. Joshi says vegetable prices have been rising and skipped the decline usually seen in winter. Besides, higher temperatures can also lead to more frequent pest attacks, he adds.

Siraj Hussain, former Agriculture Secretary and Advisor on food processing at industry body FICCI, highlights that extreme weather events have already been having an adverse impact on crop production. “This was quite visible when, due to an unusually high temperature in February 2022, wheat production was affected and India had to ban its export. Simulation studies of the Indian Council of Agricultural Research (ICAR) show that the yield of rice, wheat, maize, and other crops can go down by up to 25%,” he says.

Even where government data

has not captured the impact, clues can be found in the government’s actions. For instance, rice production, too, has been affected by erratic rains, and for confirmation, one need only look at the government’s decision to restrict export of non-basmati rice. A minimum export price was imposed on basmati rice, too, Hussain points out.

A recent report by HSBC Global Research—authored by Pranjul Bhandari, Chief Economist of India and Indonesia, and Maitreyi Das, Global Economist—points out that the heatwave of March 2022 lowered sugarcane yield by 30% while hurting the production of vegetables and oilseeds. In the heatwave of March 2024, temperatures rose to 50.5° C in some areas, leading to heat stress. The sharp rise in vegetable prices was a reflection of the resultant crop damage, it adds.

FANNING PRICE RISE

It’s not surprising then that food prices have risen sharply in the past few years.

Ramesh Chand, Member of the NITI Aayog and an agriculture economist, notes that food inflation depends on demand and supply. “In the past, inflation was cyclical, but now it has been prolonged because of several reasons. One, of course, is climate shocks. Second was the disruption in global supply,” he tells *BT*. Global wheat prices jumped after the Russia-Ukraine war began. (See Chand’s Q&A)

Government data shows that the consumer price index (CPI)-based retail inflation has been rising steadily since January 2019, when it was at 1.97%, barring a few months of decline. Retail inflation crossed the Reserve

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19.3%

dip in wheat yield in 2050 and 40% in 2080 scenarios

18-23%

decrease in *kharif* maize yields in the 2050 and 2080 scenarios

3-10%

increase in soya bean yields in 2030 and 14% in 2080 scenarios